

ORADEA, Romania

WMO: 150800

Lat: 47.0358N

Lon: 21.8958E

Elev: 450

StdP: 14.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.6	(c) 15.9	(d) 6.8	(e) 7.9	(f) 12.3	(g) 12.0	(h) 10.3	(i) 18.3	(j) 23.7	(k) 38.3	(l) 21.7	(m) 37.3	(n) 5.2	(o) 110	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.0	(c) 92.9	(d) 70.1	(e) 89.6	(f) 69.9	(g) 86.2	(h) 68.7	(i) 73.9	(j) 86.0	(k) 72.0	(l) 84.0	(m) 70.3	(n) 82.2	(o) 6.5	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	112.2	79.5	68.1	105.1	77.1	66.4	99.0	75.0	37.7	85.3	35.9	84.0	34.5	82.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.2	(b) 17.7	(c) 15.5	(e) 5.2	(f) 96.4	(g) 6.0	(h) 3.3	(i) 0.9	(j) 98.7	(k) -2.6	(l) 100.6	(m) -6.0	(n) 102.5	(o) -10.3	(p) 104.9		
(a) 20.2	(b) 17.7	(c) 15.5	(e) 4.8	(f) 76.6	(g) 6.0	(h) 2.1	(i) 0.4	(j) 78.1	(k) -3.1	(l) 79.4	(m) -6.5	(n) 80.6	(o) -10.8	(p) 82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	32.1	35.6	43.2	53.6	61.9	68.1	71.7	71.5	62.9	53.3	43.5	34.2		
	DBStd	16.00	8.79	9.39	8.40	7.07	6.44	6.22	5.67	6.07	6.77	7.89	8.50	8.74		
	HDD50	2027	557	406	238	43	3	0	0	0	1	57	229	493		
	HDD65	5144	1021	823	676	344	136	38	10	13	117	367	645	954		
	CDD50	3025	1	3	26	153	372	544	673	666	389	161	34	3		
	CDD65	667	0	0	0	3	40	132	218	214	55	5	0	0		
	CDH74	7096	0	0	0	53	470	1355	2317	2278	572	51	0	0		
	CDH80	2690	0	0	0	8	94	456	972	992	166	2	0	0		
	WSAvg	6.6	6.9	7.4	8.1	7.2	6.7	6.2	6.0	5.6	6.0	6.1	6.6	6.7		
	PrecAvg	23.6	1.3	1.3	1.3	1.8	2.2	2.9	2.9	2.0	2.2	2.0	1.9	2.0		
(15) Precipitation	PrecMax	34.6	2.6	3.9	5.6	3.3	5.3	6.3	6.6	5.3	6.0	4.3	3.5	4.7		
	PrecMin	14.1	0.5	0.0	0.2	0.3	0.6	0.5	0.3	0.0	0.4	0.1	0.0	0.0		
	PrecStd	5.4	0.6	0.9	1.3	0.7	1.1	1.5	1.4	1.4	1.4	1.2	0.9	1.1		
	PrecStd	5.4	0.6	0.9	1.3	0.7	1.1	1.5	1.4	1.4	1.4	1.2	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.7	69.4	79.2	86.1	92.5	97.0	97.1	89.8	78.9	68.3	59.2		
		MCWB	49.1	50.1	54.7	61.5	67.1	71.7	70.1	69.4	66.1	63.1	57.3	51.1		
	2%	DB	50.5	55.2	64.6	74.1	82.6	89.2	92.9	93.2	84.4	75.1	63.9	53.7		
		MCWB	46.3	48.1	52.6	58.8	65.2	70.7	70.3	69.7	64.7	61.1	54.8	48.5		
	5%	DB	47.8	51.5	60.5	70.7	79.2	85.7	89.4	89.6	80.6	71.2	59.2	49.6		
		MCWB	45.1	46.2	50.7	56.6	64.1	68.8	70.1	69.6	63.9	59.6	52.8	46.3		
	10%	DB	44.3	48.0	56.7	67.2	75.6	82.2	85.8	85.9	76.6	66.5	55.6	46.0		
		MCWB	42.4	44.5	48.6	55.2	62.6	67.9	69.2	68.6	63.1	58.1	51.4	43.8		
	0.4%	WB	49.9	51.5	56.8	63.2	70.2	75.2	76.9	76.4	69.7	65.6	58.9	52.7		
		MCDB	54.0	57.7	66.2	76.4	82.1	87.2	89.3	88.9	82.5	75.1	66.2	57.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.3	49.0	54.0	60.6	67.2	72.7	74.0	73.3	67.1	62.7	56.0	49.0		
		MCDB	50.2	54.2	62.0	71.4	78.7	84.6	85.7	86.2	78.9	71.0	61.7	53.1		
	5%	WB	44.7	46.9	51.7	58.4	65.3	70.7	72.0	71.4	65.5	60.7	53.9	46.2		
		MCDB	46.8	50.7	58.7	67.6	76.5	81.9	83.7	83.9	76.1	68.3	58.9	49.1		
	10%	WB	42.0	44.7	49.6	56.3	63.6	68.9	70.2	69.7	63.9	58.8	51.6	43.5		
		MCDB	43.7	47.5	55.8	64.9	73.6	79.4	82.2	81.9	73.6	65.7	55.3	45.5		
	5% DB	MDBR	10.5	13.1	17.2	19.9	20.5	20.7	22.0	22.3	19.9	18.5	14.0	9.7		
		MCDBR	14.5	17.9	23.3	25.7	26.2	25.7	27.8	27.5	26.0	24.2	19.0	13.8		
		MCWBR	10.6	12.0	13.6	12.9	12.2	10.6	9.9	9.3	10.2	12.1	11.9	10.1		
		MCWBR	13.0	16.1	20.9	22.4	23.6	23.0	23.9	24.4	21.9	21.0	16.9	12.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.1	11.1	13.2	12.2	12.0	10.8	10.8	10.4	10.5	11.6	11.1	9.6		
		MCWBR	10.1	11.1	13.2	12.2	12.0	10.8	10.8	10.4	10.5	11.6	11.1	9.6		
	Clear-Sky Solar Irradiance	taub	0.331	0.358	0.396	0.450	0.433	0.433	0.450	0.449	0.418	0.384	0.357	0.332		
		taud	2.324	2.280	2.208	2.080	2.194	2.228	2.187	2.187	2.240	2.317	2.340	2.352		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	234	252	259	255	265	265	258	252	249	239	222	219		
		Edh at Noon	24	31	38	48	44	43	44	43	37	30	24	21		
(44) All-Sky Solar Radiation	RadStd	RadAvg	340	606	1004	1438	1749	1904	1869	1704	1192	784	436	261		
		RadStd	43	92	140	141	167	133	158	155	135	89	48	38		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+1.62	N/A	N/A	N/A	N/A	N/A	N/A	-349	+284	+191	
(47) Regional (57 neighbors)		+1.24	N/A	N/A	+1.84	N/A	N/A	-229	-349	+203	+103	

Nomenclature: See separate page